

Project Name:

QuizBrains

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Introduction

This project is a Flashcards and Quiz System, it is designed to allow users to create and study their own flashcards on any topic. The program does not generate flashcards automatically. The user adds, edits and manages the flashcards themselves. Based on the flashcards user adds, the program automatically builds quizzes in either multiple-choice or open-answer format. The program runs on Windows, macOS and Linux as well as in common IDE-s such as VS Code and PyCharm.

Program Overview

The program provides a text-based interface that helps the user through creating their flashcards. It assists them with creating their own personal flashcards on their desired topics. When the program starts, it automatically loads all previously loaded flashcards from a data file. All user actions such as adding new flashcards or taking quizzes are saved , therefore progress is never lost. The project follows a segmented structure, meaning that the main components are separated into different python files. This makes the program easier to maintain, update and understand. Each module is responsible for a specific part of the program, such as handling files, managing flashcards, running quizzes.

Flashcard Handling Process

The central feature of the program is allowing the user to build their own personalized set of flashcards. Users can add new flashcards, edit the ones that already exist and remove the ones that they no longer want. The program checks every input in order to make sure that no input is empty or invalid, to make sure that the flashcards are usable in quizzes.

Quiz System

Once the user created the flashcards, they can start taking quizzes based on their own provided material. The program generates two types of quizzes based on their own material. The program generates two types of quizzes:

- **Multiple-choice quizzes:** where the program randomizes the order of the questions as well as the answer choices. After each quiz, the user receives their feedback and a score based on the correct answers.
- **Open-answer quizzes:** where the user types the definition of the flashcard themselves.

Every quiz attempt is also saved into a history file, including the score, the total number of questions, the percentage alongside with the timestamp. The following feature allows the users to track their progress.

File Handling

The application uses a data folder to store all information. Flashcards are saved in a file called flashcards.txt, while quiz results are stored in quiz_stats.txt. If this folder does not exist, the program creates it automatically. This ensures that the program runs smoothly on any computer, even if it is the user's first time running it.

The code is organized into several modules inside the source folder:

- **File_manager.py** manages loading and saving data
- **Flashcards_functions.py** manages flashcards operations
- **Quiz_functions.py** manages quiz logic
- **main_py** handles the main menu and connects everything

Conclusion

This project provides a simple but effective way for users to study any topic through flashcards created by the user and quizzes generated by the program. Building the system helped to strengthen topics such as file handling, input validation and user-interface design. The program is flexible, easy to make use of and fully portable across various operating systems.